

STEP 1 – Install spaCy and Load English Model

```
!pip install spacy pandas matplotlib
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
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Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1
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Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0
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Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weas
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from weasel
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->spacy)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open<8.0.0,>=5.2.1-
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en\_core\_web\_sm-3.8.0/en\_core\_web\_sm-3
```

✓ Download and installation successful

You can now load the package via `spacy.load('en_core_web_sm')`

⚠ Restart to reload dependencies

If you are in a Jupyter or Colab notebook, you may need to restart Python in order to load all the package's dependencies. You can do this by selecting the 'Restart kernel' or 'Restart runtime' option.

```
import spacy
import pandas as pd
import matplotlib.pyplot as plt
import re
from collections import Counter
```

STEP 2 – Load Twitter US Airline Sentiment Dataset

```
data = pd.read_csv("tweets dataset.csv") # upload this file in Colab
```

```
data.head()
```

	tweet_id	airline_sentiment	airline_sentiment_confidence	negativereason	negativereason_confidence
0	570306133677760513	neutral	1.0000	NaN	NaN
1	570301130888122368	positive	0.3486	NaN	0.00
2	570301083672813571	neutral	0.6837	NaN	NaN
3	570301031407624196	negative	1.0000	Bad Flight	0.70
4	570300817074462722	negative	1.0000	Can't Tell	1.00

Next steps: [Generate code with data](#) [New interactive sheet](#)

Step-3 Select needed columns & remove empty rows

```
data = data[['text', 'airline_sentiment']]
data = data.dropna()

data.head()
```

	text	airline_sentiment
0	@VirginAmerica What @dhepburn said.	neutral
1	@VirginAmerica plus you've added commercials t...	positive
2	@VirginAmerica I didn't today... Must mean I n...	neutral
3	@VirginAmerica it's really aggressive to blast...	negative
4	@VirginAmerica and it's a really big bad thing...	negative

Next steps: [Generate code with data](#) [New interactive sheet](#)

STEP-4 — Clean Tweets

```
def clean_tweet(text):
    text = re.sub(r"http\S+", "", text) # remove URLs
    text = re.sub(r"@w+", "", text) # remove mentions
    text = re.sub(r"#", "", text) # remove hashtag symbol
    text = re.sub(r"[^a-zA-Z\s]", "", text) # remove emojis & special characters
    return text.lower()
```

```
data['cleaned_text'] = data['text'].apply(clean_tweet)
```

```
data[['text', 'cleaned_text']].head()
```

	text	cleaned_text	
0	@VirginAmerica What @dhepburn said.	what said	
1	@VirginAmerica plus you've added commercials t...	plus youve added commercials to the experienc...	
2	@VirginAmerica I didn't today... Must mean I n...	i didnt today must mean i need to take anothe...	
3	@VirginAmerica it's really aggressive to blast...	its really aggressive to blast obnoxious ente...	
4	@VirginAmerica and it's a really big bad thing...	and its a really big bad thing about it	

STEP-5 – Create Cleaned Tweet Corpus

```
corpus = data['cleaned_text'].tolist()
```

```
corpus[:5]
```

```
[' what said',
 ' plus youve added commercials to the experience tacky',
 ' i didnt today must mean i need to take another trip',
 ' its really aggressive to blast obnoxious entertainment in your guests faces amp they have little recourse',
 ' and its a really big bad thing about it']
```

STEP-6 — Initialize spaCy NLP Pipeline

```
nlp = spacy.load("en_core_web_sm")
```

```
nlp
```

```
<spacy.lang.en.English at 0x7e5b5615f020>
```

STEP-7 — Create Hashtag Detector

```
from spacy.language import Language
from spacy.tokens import Doc

# Create extension to store hashtags
Doc.set_extension("hashtags", default=[], force=True)

# Register the component
@Language.component("hashtag_component")
def hashtag_component(doc):
    hashtags = []
    for token in doc:
        if token.text.startswith("#"):
            hashtags.append(token.text)
    doc._.hashtags = hashtags
    return doc

# Add component to spaCy pipeline
nlp.add_pipe("hashtag_component", last=True)
```

```
hashtag_component
def hashtag_component(doc)
```

```
<no docstring>
```

```
test_doc = nlp("I love #Indigo and #AirIndia")
test_doc._.hashtags
```

```
['#', '#']
```

STEP-8: Process all tweets using spaCy

```
docs = list(nlp.pipe(corpus))
```

```
docs[0]
```

```
what said
```

STEP-9 – Extract Lemmas and POS Tags

```
lemmas = []  
pos_tags = []  
  
for doc in docs:  
    for token in doc:  
        lemmas.append(token.lemma_)  
        pos_tags.append(token.pos_)
```

```
lemmas[:10]  
pos_tags[:10]
```

```
['SPACE',  
'PRON',  
'SPACE',  
'VERB',  
'SPACE',  
'CCONJ',  
'PRON',  
'AUX',  
'VERB',  
'NOUN']
```

STEP-10 – Extract Hashtags

```
hashtags = []  
  
for text in data['text']:  
    tags = re.findall(r"#\w+", text)  
    hashtags.extend(tags)
```

```
hashtag_freq = Counter(hashtags)
```

```
hashtag_freq.most_common(10)
```

```
[('#DestinationDragons', 75),  
( '#fail', 57),  
( '#jetblue', 35),  
( '#UnitedAirlines', 35),  
( '#customerservice', 34),  
( '#usairwaysfail', 26),  
( '#AmericanAirlines', 24),  
( '#disappointed', 22),  
( '#avgeek', 19),  
( '#badservice', 19)]
```

STEP-11 – Plot Hashtag Frequencies

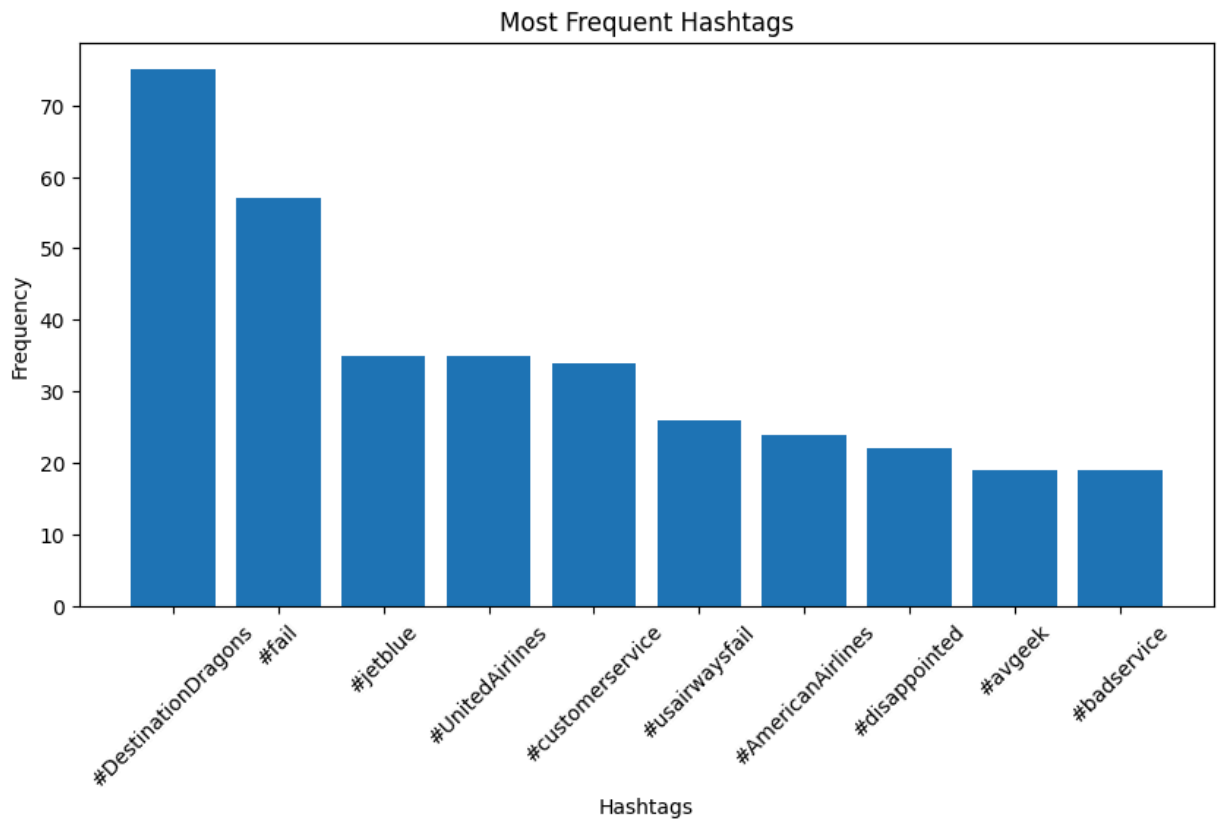
```
top_hashtags = hashtag_freq.most_common(10)
```

```

labels, values = zip(*top_hashtags)

plt.figure(figsize=(10,5))
plt.bar(labels, values)
plt.xticks(rotation=45)
plt.title("Most Frequent Hashtags")
plt.xlabel("Hashtags")
plt.ylabel("Frequency")
plt.show()

```



Step 12(a) – Filter only negative tweets

```
negative_data = data[data['airline_sentiment'] == 'negative']
```

```
neg_docs = list(nlp.pipe(negative_data['cleaned_text']))
```

```

neg_pos_tags = []

for doc in neg_docs:
    for token in doc:
        neg_pos_tags.append(token.pos_)

```

```
pos_freq = Counter(neg_pos_tags)
```

```
pos_freq
```

```

Counter({'SPACE': 15129,
        'PRON': 21006,
        'ADV': 9334,
        'ADJ': 10003,
        'PART': 7283,
        'VERB': 27196,
        'NOUN': 37487,
        'ADP': 17688,

```

```
'CCONJ': 4212,  
'DET': 10743,  
'AUX': 12871,  
'PROPN': 4816,  
'NUM': 630,  
'SCONJ': 3770,  
'INTJ': 1078,  
'X': 172,  
'PUNCT': 21,  
'SYM': 7})
```

```
plt.figure(figsize=(8,5))  
plt.bar(pos_freq.keys(), pos_freq.values())  
plt.title("POS Tag Distribution in Negative Tweets")  
plt.xlabel("POS Tags")  
plt.ylabel("Frequency")  
plt.show()
```

